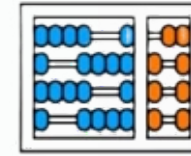




BIOLOGIA
MOLECULAR E
MORFOFUNCIONAL



Instituto de
Computação

UNIVERSIDADE ESTADUAL DE CAMPINAS

Gene Expression Networks in Prostate Cancer With and Without Obesity

Entrega 2: Differential Expression → Enrichment → Network

Course & Instructors

Course:

MO413 — Data Science & Visualization for Health (UNICAMP)

Instructors:

- Prof André Santanchè
- Murilo Vieira Geraldo

Team Members

- Cleverson Faria de Oliveira (298244)
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- Micheli Canuto de Lima (295866)

Project Details

- Repo: [GitHub](#)
- Dataset: GSE79021 (FFPE)
- Tools: RStudio, GEOquery, t-tests + volcano, g:Profiler, STRING/stringApp, Cytoscape (Leiden/Louvain), Eigenvector size

Agenda

01

Motivation & Aims

02

**Data & Preprocessing
(GSE79021)**

03

**Graph Model & Network
Strategy**

04

**Differential Expression (4
contrasts)**

05

**Functional Enrichment
(UP/DOWN)**

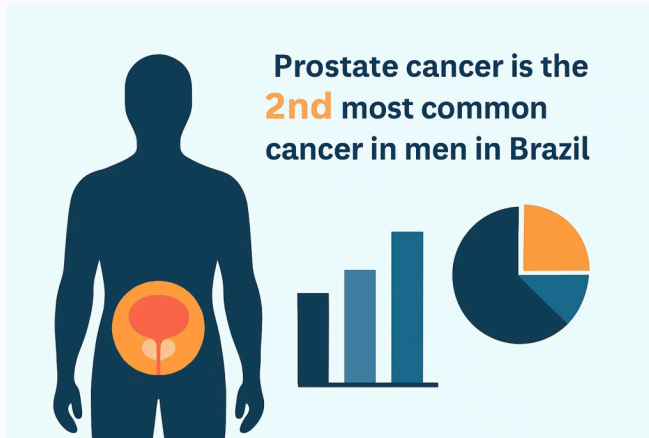
06

**PPI Network: Hubs &
Communities**

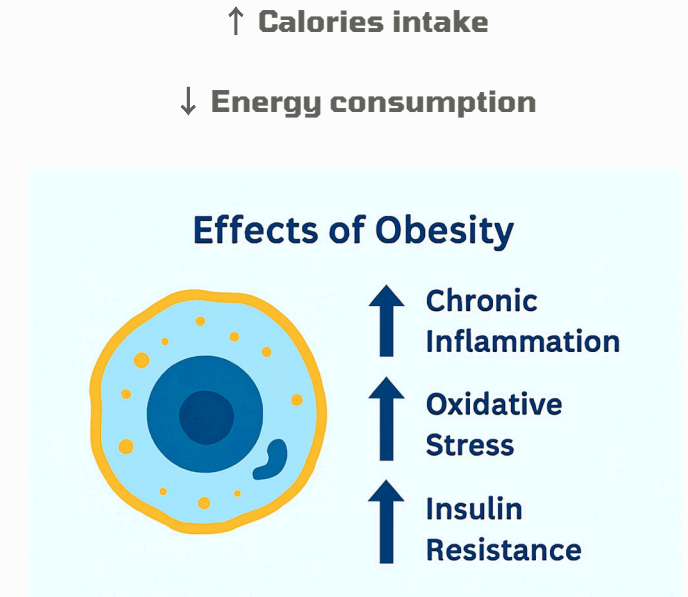
07

Interpretation, Limitations, Next Steps

Motivation & Aims



Created by Copilot AI.



Environment conducive to negatively influencing prostate tumor biology.

Motivation & Aims

Analyze gene expression data from obese and non-obese prostate cancer patients, constructing expression networks to identify gene clusters, hubs and biological pathways associated with tumor progression.

Hypotheses

- Obesity introduces a distinct expression signature in prostate tissue.
- Resulting DEGs are enriched for inflammation, metabolism, and signaling pathways.
- Within tumors, obesity adds a smaller, secondary modulation relative to the tumor–normal effect.

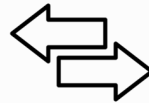
Dataset & GEO:

 Câncer de próstata

 Body Mass Index (BMI)

- Dados Sintéticos:

Perfil de expressão genética



Estado obesidade paciente

Datasets: GSE79956, GSE72291 & GSE79021

GSE79956, GSE72291

⊗ Informação pública do IMC

⊗ Sem resposta

GSE79021 & Groups

Platform: GEO (FFPE samples), SOFT parsed in R.

Tumor, non-obese (Tumor_NP):	n=143
Tumor, obese (Tumor_OB):	n=10
Normal, non-obese (Normal_NP):	n=46
Normal, obese (Normal_OB):	n=3 (very small; exploratory only)

Dataset & Groups (GSE79021)

Platform: GEO (FFPE samples), SOFT parsed in R.

Groups and counts:

Tumor, non-obese (Tumor_NP):	n=143
Tumor, obese (Tumor_OB):	n=10
Normal, non-obese (Normal_NP):	n=46
Normal, obese (Normal_OB):	n=3 (very small; exploratory only)

Methods (compact)



Parse SOFT → sample metadata (group, BMI) + expression table.



Build long → wide matrix; subset by group.



Per-gene Welch t-tests (two-group), compute log2FC & $-\log_{10}(p)$.



Volcano thresholding: $|\log_2FC| \geq 0.30$; $p < 0.05$ (unadjusted).

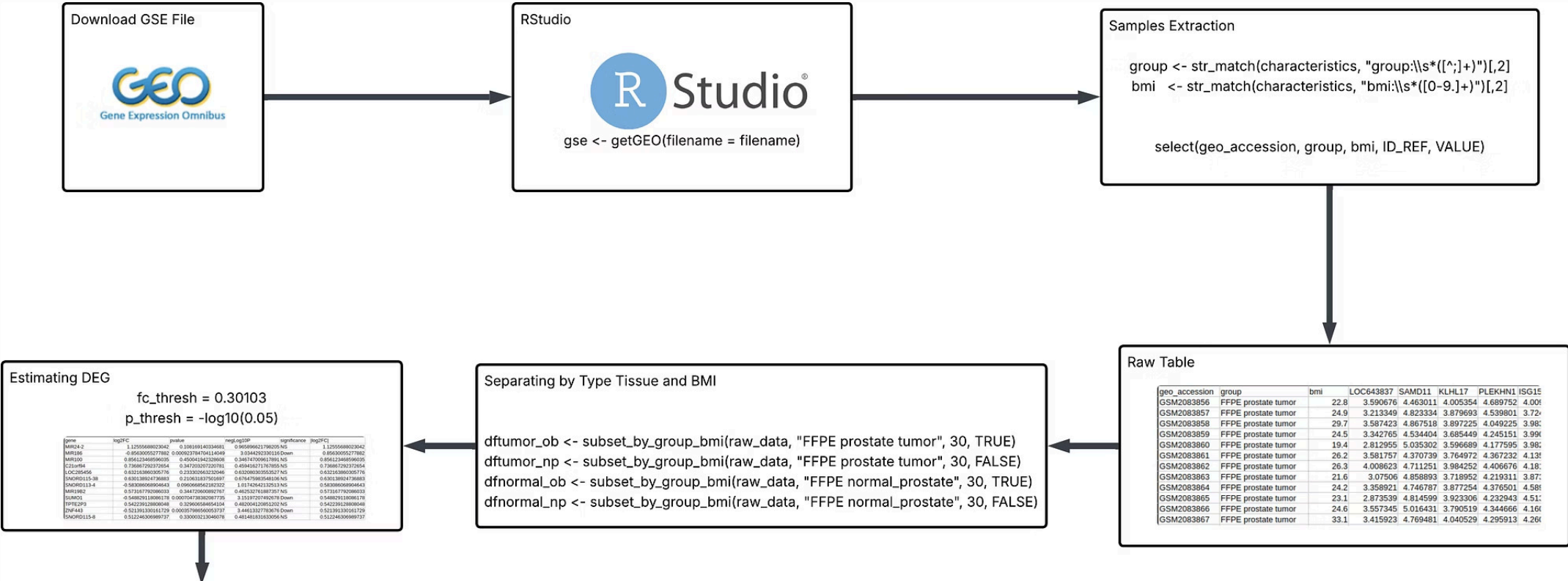


Export top DEG lists (Up/Down) → STRING seeds; create Cytoscape nodes with DE class.



Plan next: limma (eBayes), FDR control, pathway & network analysis.

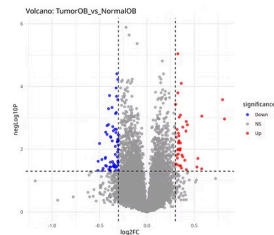
Code Workflow



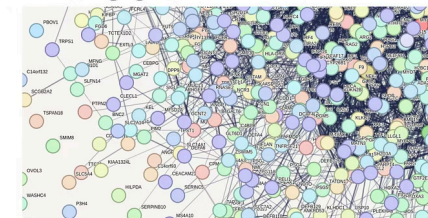
Code Workflow

Generates Volcano Plots

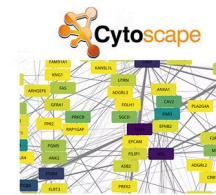
```
deg$significance <- "NS"  
deg$significance[deg$log2FC > fc_thresh & deg$negLog10P > p_thresh] <- "Up"  
deg$significance[deg$log2FC < -fc_thresh & deg$negLog10P > p_thresh] <- "Down"
```



String WebSite



Cytoscape



Results — DEG Summary

Cutoffs: $|\log_2FC| \geq 0.30$ and $p < 0.05$

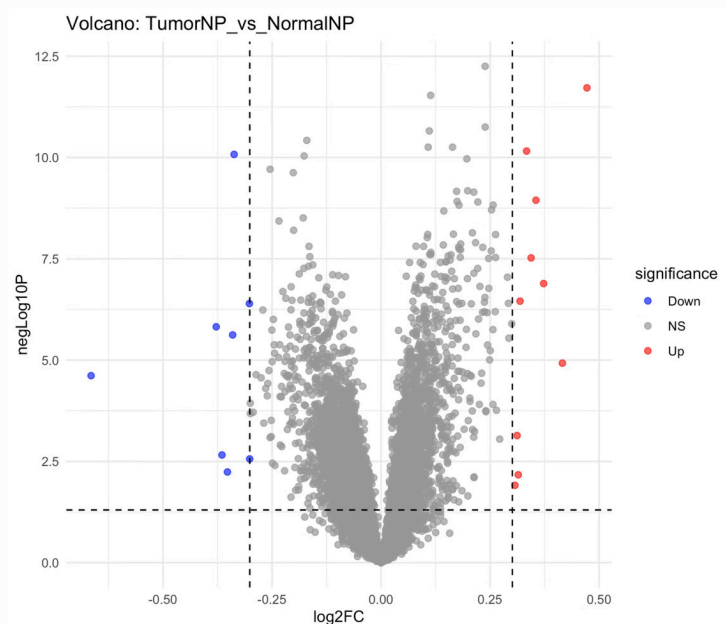
TumorNP vs NormalNP:	10 up / 8 down
TumorOB vs NormalOB:	41 up / 64 down (Normal_OB n=3)
TumorOB vs TumorNP:	3 up / 9 down
NormalOB vs NormalNP:	12 up / 27 down

Results — DEG Summary

Volcano — TumorNP vs NormalNP

Interpretation:

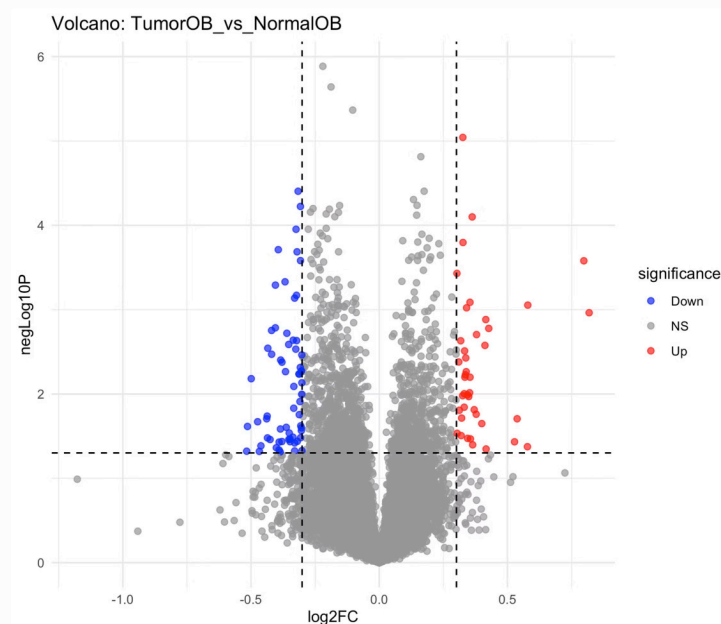
- Clear but modest tumor–normal signal in non-obese; many genes sit near the fold-change boundary.
- Likely underestimates true effect without shrinkage (no limma yet).



Volcano — TumorOB vs NormalOB

Interpretation:

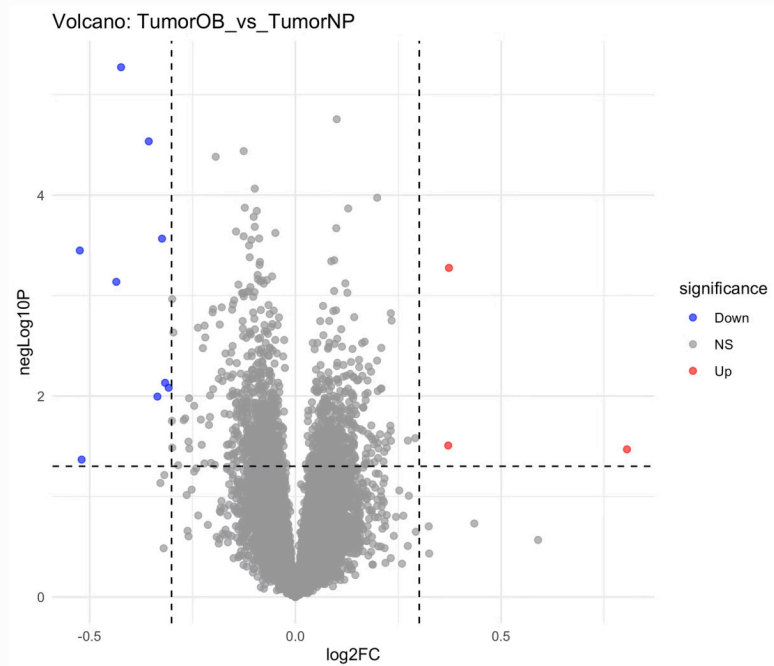
- Larger DEG set (41 up / 64 down), suggesting strong tumor–normal separation in obese.
- Caution: Normal_OB has n=3, making p-values unstable; treat as exploratory.



Volcano – TumorOB vs TumorNP

Interpretation:

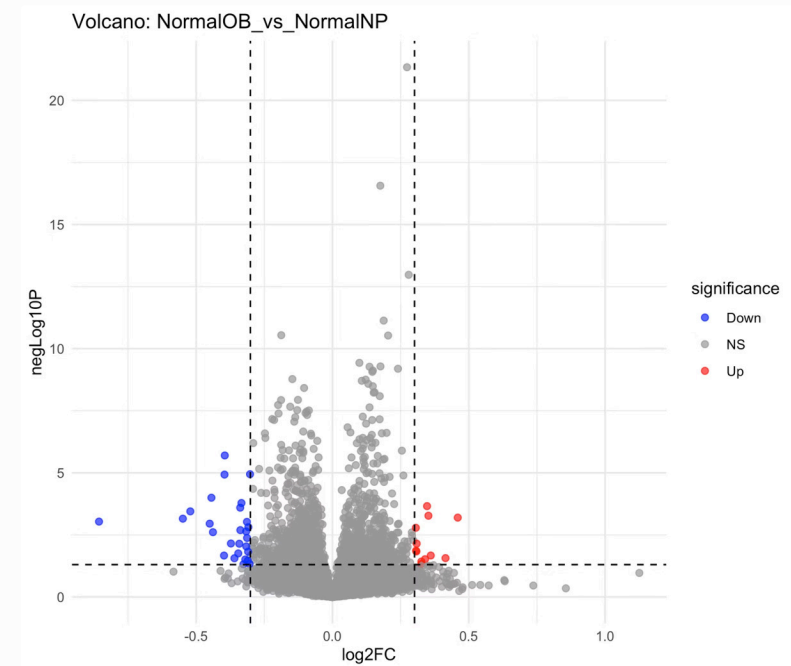
- Only 3 up / 9 down → small obesity effect within tumor compared to tumor-normal.
- Likely underpowered (10 vs 143). Consider moderated statistics.



Volcano – NormalOB vs NormalNP

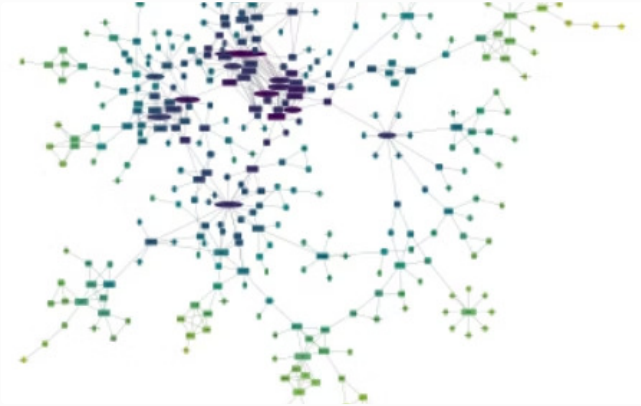
Interpretation:

- 12 up / 27 down indicates obesity shifts expression in normal prostate, consistent with metabolic/inflammatory stressors.
- Provides plausible mechanisms that may prime the microenvironment.



Cytoscape results

Tumor OB vs Tumor NP



Protein-Protein Interaction Networks

This comparison has 380 nodes and 675 edges.



Hub Genes and Network Centrality

The hubs identified include PIK3CB, ITGB3, RBBP4, H2BC5, H2AC14, and CAV1 similar to those found in the Tumor NP network.

Avg. number of neighbors: 3.384
Network diameter: 21
Network radius: 11
Characteristic path length: 7.608
Clustering coefficient: 0.309
Network density: 0.018
Network heterogeneity: 1.026
Network centralization: 0.152

Analyze Network

- Nodes: **380**
- Edges: **675**
- Density: **0.009**
- Clustering: **0.359**
- Heterogeneity: **0.940**
- Centralization: **0.086**

Global Comparison

Comparison	Nodes	Density	Clustering	Heterogeneity	Centralization	General Interpretation
TumorNP vs NormalNP	621	0.007	0.262	0.985	0.045	Broad, distributed network, with several moderate hubs.
NormalOB vs NormalNP	308	0.013	0.393	0.898	0.092	More clustered, modular, stable network.
TumorOB vs NormalOB	185	0.018	0.309	1.026	0.152	Hierarchical network, dependent on few hubs.
TumorOB vs TumorNP	380	0.009	0.359	0.940	0.086	Balanced, modular, adaptive network.

Key Hub Genes Across Contrasts

Further analysis identified key hub genes in each comparison, indicating their central roles within the protein-protein interaction networks and potential impact on disease progression and obesity-related effects.

1

TumorNP vs NormalNP

- **PIK3CB (36):** Dominant node, cellular signaling regulator.
- **ITGB3 (19):** Highly connected, links cell adhesion modules.
- **RBBP4 (17):** Connects epigenetic regulation modules.
- **H2BC5 (16):** Intermediate node, bridge between histone complexes.
- **H2AC14 (14):** Connected to central nodes, structural gene.
- **CAV1 (14):** High closeness centrality, involved in membrane organization.

2

NormalOB vs NormalNP

- **COL1A1/COL3A1/COL4A1:** Structural genes, highly connected in extracellular matrix networks.
- **VCAN:** Intermediate-high degree node, acts as a connector between groups.
- **CDH1:** Node with high betweenness centrality.

3

TumorOB vs NormalOB

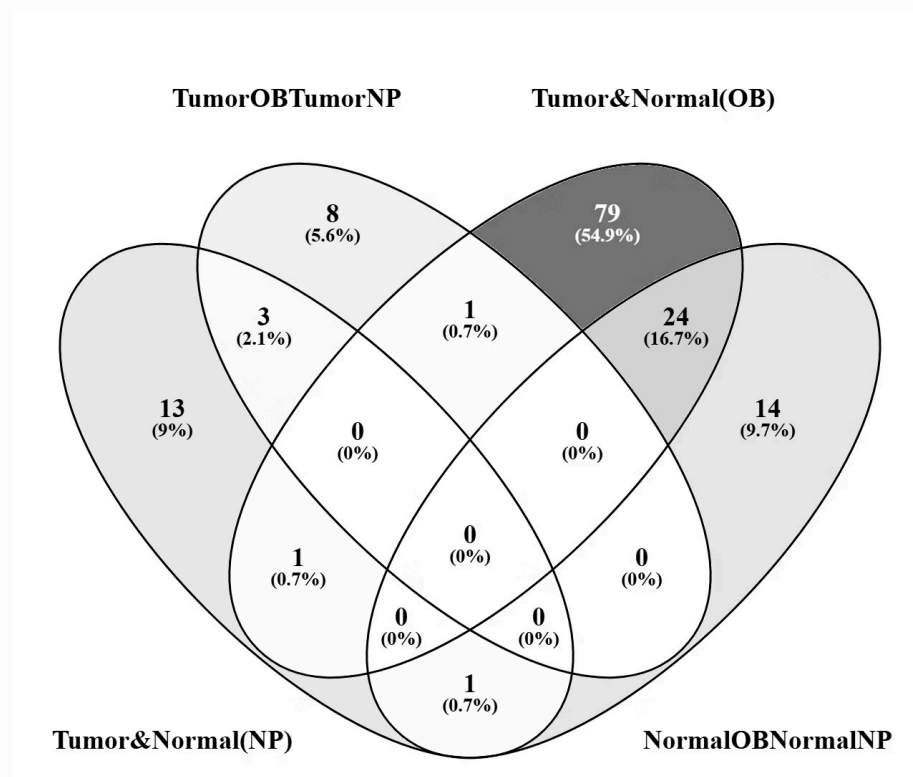
- **H4C6 (31):** Largest hub with extremely high connectivity.
- **GNA11 (17):** Signaling node, high betweenness.
- **H2AC14 (16):** Histone with broad connections.
- **H2BC5 (14):** Participates in the structural core of the network.
- **CBX3 (13):** Connects different sub-modules, acts as an intermediate node.
- **RBBP4 (12):** Bridge node with high closeness centrality.

4

TumorOB vs TumorNP

- **PIK3CB (36):** Primary control node, consistent with TumorNP.
- **ITGB3 (19):** Second relevant hub, connects adhesion-related modules.
- **RBBP4 (17):** Node with high betweenness centrality.
- **H2BC5 (16):** Structural histone with multiple connections.
- **H2AC14 (14):** Structural bridge node.
- **CAV1 (14):** Regulates several peripheral sub-modules.

Venn's Diagrams



Oliveros, J.C. (2007-2015) Venny. An interactive tool for comparing lists with Venn's diagrams. <https://bioinfogp.cnb.csic.es/tools/venny/index.html>

Venn Diagram Summary

79 genes (54.9%) → exclusive to Tumor vs Normal (OB)

13 genes (9%) → exclusive to Tumor vs Normal (NP)

8 genes (5.6%) → exclusive to Tumor (OB) vs Tumor (NP)

14 genes (9.7%) → exclusive to Normal (OB) vs Normal (NP)

Observations:

The Venn diagram analysis reveals no overlap between differentially expressed genes in normal and tumor tissues from obese versus non-obese individuals, suggesting that obesity-induced transcriptional changes are tissue-context dependent and may involve distinct molecular mechanisms in normal versus malignant prostate microenvironments.

Tumor ob _ Tumor np (DAVID)

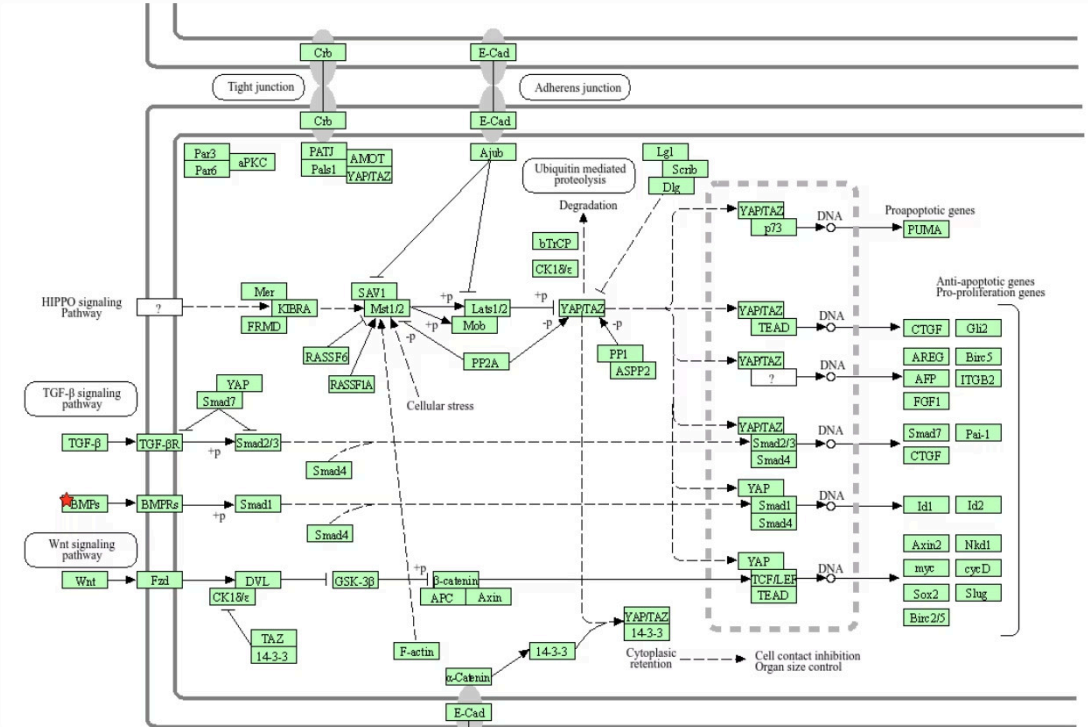
The common genes identified (**BMP5, DNAH5, KCNH8, VCAN, TDRD1, and OR51G2**) were associated with several disease classes, mainly **metabolic, cancer, and cardiovascular** categories. Other associations included **reproductive, neurological, immune, and psychological** terms. Overall, **metabolic and cancer-related pathways were the most frequent**, suggesting a strong link between these genes and processes involved in metabolism and tumor development.

Tumor ob _ Tumor np (DAVID - BMP5)

Gene compartilhado entre o grupos Tumor OB vs Tumor Np & Tumor&Normal (NP)

Up regulated

BMP5 interacts with the Hippo, TGF- β , and Wnt pathways, regulating cell growth and apoptosis. Its dysregulation may promote prostate cancer progression by enhancing proliferation and inhibiting cell death.



Functional Annotation Table

Summary of Four Contrasts Analysis



Key Findings from the four comparisons:

- **TumorNP vs NormalNP:** 18 DEGs (10 up/8 down) - modest but clear tumor signature.
- **TumorOB vs NormalOB:** 105 DEGs (41 up/64 down) - strongest signal, but limited by small Normal_OB sample size.
- **TumorOB vs TumorNP:** 12 DEGs (3 up/9 down) - subtle obesity effect within tumors.
- **NormalOB vs NormalNP:** 39 DEGs (12 up/27 down) - obesity impact on normal tissue.



Network Topology Insights:

- Different network architectures across comparisons.
- Hub gene conservation (PIK3CB, ITGB3, RBBP4) in tumor-related contrasts.
- Tissue-specific hubs in normal comparisons (COL genes, VCAN).



Biological Implications:

- Obesity creates distinct molecular signatures in normal prostate tissue.
- Tumor-normal differences dominate over obesity effects.
- No gene overlap between tissue types suggests context-dependent mechanisms.

Limitations

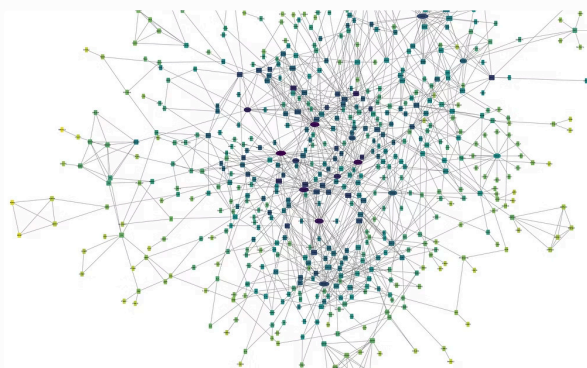
- Unbalanced groups (Normal_OB n=3; Tumor_NP n=143) affected variance and p-value sensitivity.
- Multiple testing correction (FDR) not yet applied.
- FFPE data may introduce technical variability.
- BMI used as binary split; continuous BMI analysis needed.

Thank you!

Any questions?

Cytoscape results

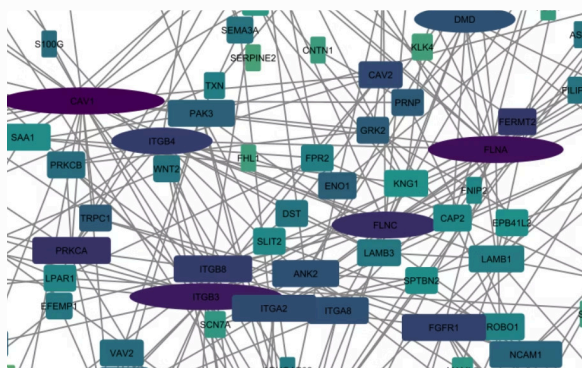
Tumor NP vs Normal NP



Protein-Protein Interaction Networks

This network represents the interaction structure between differentially expressed genes in the Tumor NP and Normal NP groups.

The network contains 621 nodes (genes) and 1264 edges (interactions).



Hub Genes and Network Centrality

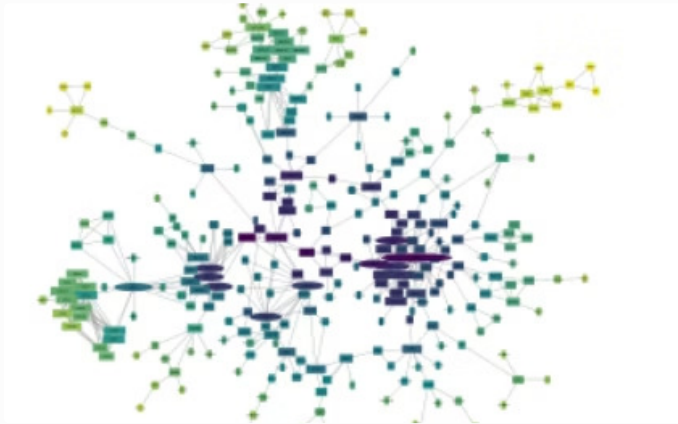
The highlighted nodes in the second figure represent hub genes, identified through high degree and betweenness centrality values. These hubs such as PIK3CB, ITGB3, RBBP4, H2BC5, and CAV1 – serve as central connectors that link multiple modules within the network.

Avg. number of neighbors: 4.071
Network diameter: 13
Network radius: 7
Characteristic path length: 5.248
Clustering coefficient: 0.262
Network density: 0.007
Network heterogeneity: 0.985
Network centralization: 0.045

Analyze Network

- Nodes: **621**
- Edges: **1264**
- Densidad: **0.007**
- Clustering: **0.262**
- Heterogeneidad: **0.985**
- Centralización: **0.045**

Normal OB vs Normal NP



Protein-Protein Interaction Networks

This network connects 308 nodes with 595 edges, showing interactions under normal conditions.



Hub Genes and Network Centrality

The hubs are visualized as larger or darker nodes.

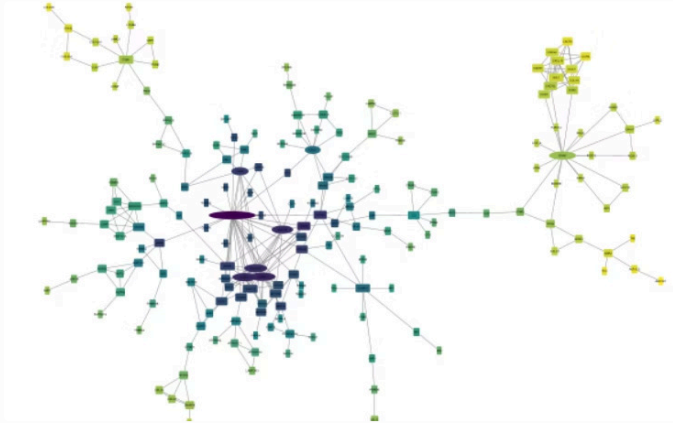
They correspond to locally central genes such as COL1A1, COL3A1, VCAN, and CDH1, which connect smaller neighborhoods.

Avg. number of neighbors: 3.553
Network diameter: 21
Network radius: 13
Characteristic path length: 8.149
Clustering coefficient: 0.359
Network density: 0.009
Network heterogeneity: 0.940
Network centralization: 0.086

Analyze Network

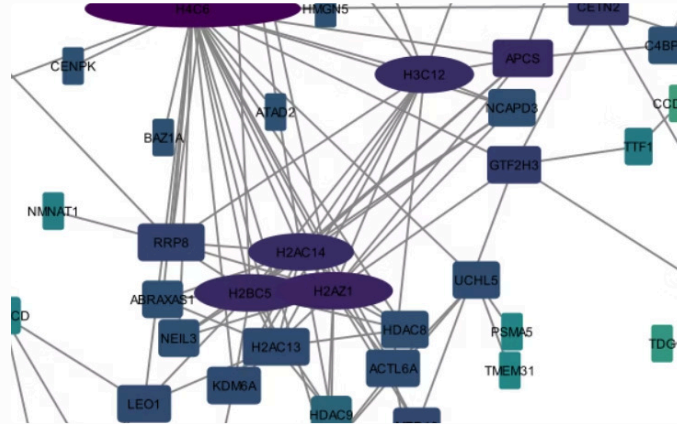
- Nodes: **308**
- Edges: **595**
- Densidad: **0.013**
- Clustering: **0.393**
- Heterogeneidad: **0.898**
- Centralización: **0.092**

Tumor OB vs Normal OB



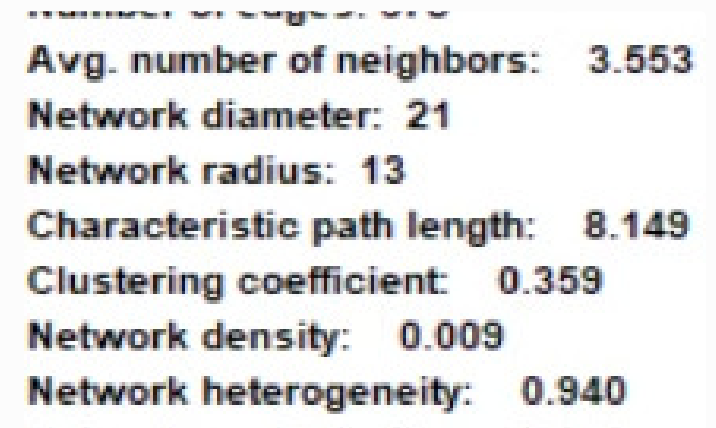
Protein-Protein Interaction Networks

This network contains 185 nodes and 313 edges, making it the most compact structure among the comparisons.



Hub Genes and Network Centrality

The main hubs are H4C6, GNA11, H2AC14, H2BC5, CBX3, and RBBP4. Among them, H4C6 stands out as the most connected gene (degree = 31), acting as the dominant hub that links most modules.



Analyze Network

- Nodes: **185**
- Edges: **313**
- Densidad: **0.018**
- Clustering: **0.309**
- Heterogeneidad: **1.026**
- Centralización: **0.152**

Next Steps (Entrega 2 → 3)

Refined Statistics

Re-fit with limma (voom if RNA-seq/appropriate normalization); report FDR and effect sizes.



Functional Enrichment

Gene set enrichment (Up/Down separately) using g:Profiler/Reactome/KEGG for each contrast.

Network Analysis

Build STRING PPI networks from top DEGs; export to Cytoscape. Node size = eigenvector centrality, Node color = Up/Down, cluster with Leiden.



Sensitivity & Validation

Sensitivity: stricter cutoffs (e.g., $|\log_2FC| \geq 0.5$), add covariates if available.
Validation: test DEG signatures against TCGA-PRAD (Gleason, stage, PFS).

Reproducibility

- Code: R (GEOquery, dplyr, ggplot2, plotly); script parses SOFT, generates subsets & volcanoes, exports STRING/Cytoscape inputs.
- Thresholds exactly as shown on plots: $|\log_2FC| \geq 0.30$; $p < 0.05$.
- Artifacts produced:
 - Volcano_*.png (4 plots)
 - DEG_*.csv (per contrast)
 - STRING_genes_UP/DOWN_*.txt
 - NODES_for_Cytoscape_*.csv

Takeaway

Obesity's Role

Obesity reshapes normal prostate expression, consistent with metabolic/inflammatory mechanisms.

Future Focus

We'll stabilize results with limma + FDR, map pathways, and build PPI/cluster views to pinpoint candidate hubs for validation in TCGA-PRAD.

Tumor Dominance

Tumor-normal differences remain primary; within-tumor obesity effects are subtle in this dataset.